Prostatectomy for prostate cancer

Some people with prostate cancer have an operation to remove the whole prostate (a radical prostatectomy). The aim is to remove all the cancer.

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What is a radical prostatectomy?

Surgery to remove the prostate to treat prostate cancer is called a radical prostatectomy. There are different types of radical prostatectomy. The surgery is done by a urologist or urology surgeon who specialises in treating prostate cancer and performs this type of surgery often. Not all urologists do surgery for prostate cancer.

The aim of a radical prostatectomy is to remove all the cancer by removing the prostate.

Radical prostatectomy for early prostate cancer

A radical prostatectomy can be done when the cancer is within the prostate. It is sometimes done when the cancer has spread just outside the prostate.

It is a major operation. It may not be suitable for everyone. It depends on your general health, or any other health condition that could increase the risks of surgery. The urologist and specialist nurse can talk to you about other treatments that may also be effective. For example, these may include active surveillance, radiotherapy or brachytherapy. They may refer you to an oncologist (cancer doctor) to talk about other treatment options.

Radical prostatectomy for locally advanced prostate cancer

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A radical prostatectomy is not always suitable when the cancer is locally advanced. It may not be possible to remove all the cancer cells that have spread outside the prostate. The urologist and specialist nurse can talk to you about whether a radical prostatectomy would be suitable for you, or whether other treatment such as radiotherapy may be best for you. They may refer you to an oncologist (cancer doctor) to talk about other treatment options.

There is also surgery that can help with symptoms. A transurethral resection of the prostate (TURP) does not treat the cancer, but it can help urinary symptoms, such as difficulty passing urine (peeing). It is sometimes done before other treatments such as radiotherapy. This is because radiotherapy can make urinary symptoms worse during and for a while after treatment.

Before the operation, the urologist and specialist nurse will explain what will happen and tell you about the possible side effects and risks. You will be given written information to take away. It is important you have all the information you need to decide about having surgery.

Types of prostatectomy

There are different ways of doing a radical prostatectomy. Your surgeon will explain the type of surgery you will have. It is usually laparoscopic (keyhole) surgery. This is often robotic-assisted laparoscopic surgery. Occasionally, it may be open surgery.

During a radical prostatectomy, the surgeon frees the prostate gland from the bladder and urethra so that it can be removed. They rejoin the urethra to the bladder using dissolvable stitches. They also remove the seminal vesicle, the part of the body that helps make semen. They may also remove the lymph nodes close to the prostate and check them for cancer cells. This depends on your risk of having cancer in the lymph nodes.

A radical prostatectomy is usually done under a general anaesthetic, which means you are not awake when you have the procedure. But it can also be done under a spinal anaesthetic. If you have a spinal anaesthetic, you will be given a sedative to help you relax and feel sleepy.

You will meet with an anaesthetist to talk about which anaesthetic you will have, and pain control after the surgery.

We have more information about the healthcare professionals you might meet when having surgery, and about types of anaesthetic.

Laparoscopic (keyhole) radical prostatectomy

In this type of surgery, the surgeon does not need to make a large cut. Instead, they remove the prostate using 5 or 6 small cuts in the tummy area (abdomen). Each cut is about 1cm long.

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The surgeon puts a small tube with a light and camera on the end through 1 of the cuts. This is called a laparoscope. The camera shows an image of the prostate on a screen. The surgeon uses small, specially designed equipment to cut away the prostate from the bladder and urethra. They remove the prostate through 1 of the small cuts.

Robotic-assisted laparoscopic radical prostatectomy

Robotic-assisted surgery is when a laparoscopic radical prostatectomy is assisted by a machine. Instead of the surgeon holding the laparoscope and the equipment, they are attached to robotic arms. The surgeon controls the robotic arms, which they move very precisely. The aim of robotic-assisted surgery is to help prevent damage to the nerves in the area. These are the nerves that control your bladder and erections.

Surgeons need special training to do robotic-assisted surgery, and not all hospitals have a robot. Your urologist can tell you whether this surgery is suitable for you, and whether you need to go to another hospital.

Open radical prostatectomy

During open surgery, the surgeon makes 1 larger cut in your abdomen. It goes downwards from your belly button. They remove the prostate and seminal vesicles through this cut, and lymph nodes if needed. Sometimes a cut can be made in the area between the scrotum and anus instead. This is called the perineum.

Before a radical prostatectomy

Before a radical prostatectomy, the surgeon and specialist nurse will explain the possible risks and side effects.

Before surgery, your healthcare team may advise you on what you can do to be fitter for surgery, such as stopping smoking, being more physically active or eating healthily. This is called prehabilitation.

Some hospitals might have a prehabilitation service. But if your hospital does not, your prehabilitation may involve specific help and support to prepare for surgery from your surgeon, specialist nurse and other professionals in the multidisciplinary team (MDT).

Your surgeon and specialist nurse may also talk about things you can do to reduce the time you spend in hospital and help you to recover as quickly as possible. This is called enhanced recovery.

Before your surgery, you may also have a pre-operative assessment. This may involve having some tests to check your general health and fitness. You can also learn more about how to prepare for your surgery at this appointment. You may be invited to a group education session about having a radical prostatectomy.

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You will be encouraged to start pelvic floor exercises to strengthen the pelvic floor muscles. You can do these at home. This can help reduce urinary leakage (incontinence) after the surgery.

If you think you might need help at home after surgery, tell your nurse when you go into hospital. They can talk to you about the support that is available.

Going into hospital

You usually go into hospital on the day of your surgery, or the day before. When you arrive, the nurses on the ward will explain what to expect and give you some information. They will give you an identity bracelet with your details on it. You might find you meet different healthcare professionals and you may be asked the same questions each time. This is just to check everything is correct at each stage of your care.

Ask questions if you are unsure about anything. The nurses would prefer you to ask than be worried.

If you are going into hospital on the day of your surgery, you may be asked not to eat or drink anything for a few hours. This is sometimes called nil-by-mouth. If you have diabetes, make sure the doctors and nurses know about this.

Follow any advice you are given at the pre-operative assessment about taking your usual medicines. You can usually still take these with a sip of water, but you should check with the doctor and nurses.

Your nurse will give you compression stockings to put on before surgery. These are called TED stockings. You will also have to wear these for a period of time afterwards. Compression stockings reduce the risk of getting a blood clot in your legs. This is called deep vein thrombosis or DVT.

Compression socks and stockings need to be the right size with no folds or wrinkles to work effectively. Your nurse will measure you and then check they fit and are on properly.

After a radical prostatectomy

If you are having a general anaesthetic, your surgeon, anaesthetist or nurses will explain what to expect when you wake up after surgery. You are usually moved to the recovery room in the operating department to be monitored until you are awake. You are then moved back to the ward a short time later.

You might feel drowsy at first if you had a general anaesthetic. You may notice your face is a bit puffy. This is because your head is tipped downwards during the surgery.

Immediately after surgery, the nurses can help you with washing and going to the toilet. But they will encourage you to be up and moving around as soon as possible. They can help you move safely with your catheter. You may feel anxious about this, but moving around prevents complications and helps your recovery.

You will usually have a drip into a vein in your arm. This is called an intravenous infusion. It will stay in for a few hours after your operation, until you are eating and drinking again.

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Wound

After open surgery, you will either have a wound on your tummy area (abdomen) or perineum. This is the area between the scrotum and the anus. After laparoscopic surgery, you will have 5 or 6 small wounds in the abdomen.

If you had open surgery, you may also have a small tube going into the wound. This is called a drain. It helps to remove any fluid that is collecting there. It is usually removed after a few days.

Before you go home, the nurses on the ward will advise you on how to take care of your wound. They will advise you on what to look out for and who to contact if you have any concerns.

Pain

After surgery, you will have some bruising or swelling around the cuts and also in your scrotum.

You may have some pain or discomfort after surgery. Painkillers will help with this. In hospital, you may have pain medicines through a drip in your vein, or by injection. This will then be replaced with painkiller tablets. Tell the nurses on the ward if you are still in pain. They will give you tablets to take home with you and explain how to take them.

You might have some discomfort for a couple of weeks, particularly when you walk. Taking regular painkillers should help. Talk to your GP, specialist nurse or pharmacist if you are still getting pain.

Urinary catheter

During surgery, you will have a tube put in to drain urine from your bladder. This is called a catheter. The catheter keeps your bladder empty while the urethra heals. Try to drink plenty of fluids to help keep the catheter draining well.

The nurses on the ward will show you how to look after your catheter before you go home. They will give you spare bags to take home. If you need more support, they can arrange for a district or community nurse to visit you at home.

The nurses will give you details of who to contact if you have any problems or concerns about your catheter. The catheter is usually removed 1 to 2 weeks after surgery. The nurses may give you a clinic appointment date to have this done. Ask the nurses on the ward before you go, or call the number you have been given if you are not sure about anything to do with your catheter.

You will leak urine for a while after the catheter is removed. This is expected and should slowly improve. You will need to wear a pad in your underwear for a while, but the need for pads should lessen. Your surgeon and specialist nurse can talk to you about what to expect.

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Preventing blood clots

After your surgery, the nurses will encourage you to be up and moving around. This can help prevent complications such as blood clots in the veins in the legs. You will continue to wear your compression stockings for a while. Your nurse can explain how long to wear them for. They can talk to you about how to wash and care for them. They may give you a spare pair to take home.

The nurse may give you an injection under the skin to help prevent blood clots. When you go home, you are usually given more of the small injections. You usually have these for 28 days. The nurses can show you or your carer how to give them. They may organise for a district or community nurse to give them instead.

Going home

If you had laparoscopic or robotic surgery, you usually go home after 1 to 2 days. After open surgery, it is usually between 3 and 7 days. Before you leave hospital, your hospital doctor will send a letter called a discharge summary to your GP. The letter is usually sent electronically so that your GP gets it within 24 hours of your discharge. You are usually given a copy to take home with you. It will explain your diagnosis, what type of surgery you had, any medication you are taking and about your follow-up care.

If you need a district or community nurse, the ward nurses will organise this before you go home. You should also be given 24-hour telephone numbers for your hospital team in case you need to contact them.

You can usually get back to normal activities 4 to 12 weeks after surgery, depending on the operation you had. If you had robotic-assisted surgery, you usually recover faster.

It is important to remember that even if you have small wounds, you have still had major surgery. You will feel tired, so try to get plenty of rest and eat well. Do some light exercise, such as walking, to build up your energy. You can slowly increase the amount you do.

Your surgeon, specialist nurse or nurses on the ward can give you advice on when you can start doing things such as driving or returning to work. They can also give you advice on when you can have sex again.

Follow-up after surgery

Your PSA level will be checked about 6 to 8 weeks after surgery. This can help your doctors to know whether they have removed all the cancer.

You will have a clinic appointment to meet with the surgeon. They will check that your wound is healing properly and tell you about:

- results from the tissue removed during surgery (pathology)

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- the stage of the cancer
- any further cancer treatment you may need
- your recovery after surgery.

Side effects of a radical prostatectomy

There are side effects after a radical prostatectomy. Some side effects improve over time, but some are permanent.

Erection problems

Surgery to the prostate can cause problems getting or keeping an erection. This is called erectile dysfunction (ED). This is caused by damage to the nerves and blood vessels close to the prostate. These are the nerves and blood supply that help you get an erection.

Surgeons can operate in a way that tries to protect these nerves or blood vessels. This is called a nerve-sparing technique. It is only possible if the cancer has not spread to the edges of the prostate. During surgery, if the surgeon thinks there is cancer in the nerves or surrounding area, they remove some or all the nerves.

Whether you will have problems getting an erection after a nerve-sparing operation depends on different factors. For example, it may depend on your age and whether:

- you had erection problems before surgery
- you have any other medical conditions, such as diabetes
- you are taking medicines for high blood pressure
- you have had surgery called a transurethral resection of the prostate (TURP)
- the surgeon was able to spare some or all of the nerves.

You can ask your surgeon about your risk of ED. Your ability to have an erection may slowly return after surgery. But this may take 1 or 2 years. It is less likely to return if you have further treatment after surgery, such as hormonal therapy or radiotherapy.

You may be offered penile rehabilitation using different ED treatments soon after surgery. This is called an ED recovery package.

Ejaculation

When you have your prostate removed, you can still have an orgasm even without an erection, but there will be no ejaculation. This is called a dry ejaculation or dry orgasm. It may cause some discomfort at first, but this usually improves with time. You may pass a small amount of urine when you orgasm. Talk to your urologist or specialist nurse if you are having this problem. They may be able to give you some advice.

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Fertility

Having your prostate removed will affect your fertility. This is your ability to make someone pregnant. The prostate and seminal vesicles produce semen, which is normally mixed with sperm from the testicles. Removing the prostate means you will not be able to ejaculate any more.

It is important to talk to your urologist or specialist nurse about fertility before treatment. It may be possible to store sperm before your surgery.

Bladder problems

It is usual to have some leaking from the bladder when the catheter is first removed. This is called urinary incontinence. It usually improves within a few weeks or months after surgery. You will be encouraged to do pelvic floor exercises to strengthen the pelvic floor muscles. You will need to wear a pad to manage the incontinence, but the need for pads should lessen.

Sometimes, you may have some incontinence when you cough, sneeze or exercise. This is called stress incontinence. It is rare to be completely incontinent, but it can happen. If you are having problems, talk to your GP, urologist or specialist nurse. They can refer you to a continence specialist who can give you treatment, information and support to manage your symptoms. If these do not help, you may be able to have surgery to improve your symptoms.

Another less common side effect can be scarring to the urethra or the entrance of the bladder. This is called the bladder neck. Scar tissue can narrow the bladder neck or the urethra and make passing urine difficult. This causes urine to build up in the bladder and overflow, causing you to leak urine. This can usually be treated with a small operation that opens up the bladder neck or the urethra. If you are having problems passing urine after surgery, talk to your GP, urologist or specialist nurse.

Macmillan toilet card

If you need to use a toilet urgently, Macmillan has a toilet card that you can use in places such as shops, offices, cafés and pubs. You can use it during or after treatment. We hope it allows you to get access to a toilet without any awkward questions. But we cannot guarantee that it will work everywhere.

Benefits and disadvantages of a prostatectomy for early prostate cancer

Benefits

- If the cancer has not spread outside the prostate, removing it may cure the cancer and you will not need any more treatment.
- If the cancer comes back, you will still be able to have further treatment.
- If you had urinary symptoms before surgery, these may improve after surgery.

Disadvantages

- There is a small risk of problems after the surgery, such as bleeding or infection.

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- Surgery may cause long term problems with erectile dysfunction and incontinence.
- Removing your prostate means you will no longer be able to make someone pregnant naturally.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Below is a sample of the sources used in our prostate cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

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The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

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Get help

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